

Robot Framework

Open-source keyword-driven framework extensible through libraries for Web, API, desktop and other domains.

What it is and when to use it

Robot Framework uses tabular syntax and keywords, supporting readable and reusable high-level automation.

Prerequisites

Install Python 3 and pip. On Windows add Python to PATH. Verify `python --version` and `python -m pip --version`.

Recommended virtual environment

Create a project folder, run `python -m venv .venv` and activate it. Use `.venv\Scripts\Activate.ps1` on PowerShell or `source .venv/bin/activate` on macOS/Linux.

Step-by-step installation

Run `python -m pip install robotframework` and verify `robot --version`. Add a suitable Web library such as SeleniumLibrary or Browser Library when needed.

First suite

Create a `.robot` file with Settings, Test Cases and Keywords and run `robot path/to/suite.robot`. Review `output.xml`, `log.html` and `report.html`.

Keywords and organization

Design business-oriented keywords and keep resources, variables and libraries predictable.

Libraries

Choose libraries according to the tested technology and project maturity.

CI/CD and reports

Install requirements, run robot and publish log/report/evidence artifacts.

Troubleshooting

Check PATH, active virtual environment, Python version and whether packages were installed into the correct interpreter.

QA tip

Use the tool within a quality strategy: automate meaningful risks, keep the suite observable and treat automation failures as production-code issues.

Official resources

Site oficial: <https://robotframework.org/>

Download / instalação: <https://robotframework.org/robotframework/latest/RobotFrameworkUserGuide>

Documentação: <https://docs.robotframework.org/>